IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-089

1. Topic (12 words max)

Data-Driven Analysis of University Past Papers and Lecture Content for Curriculum Optimization

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Universities frequently face challenges in objectively evaluating the alignment between their curriculum and assessment of content. This misalignment creates a significant gap, causing students to inefficiently allocate revision efforts, spending excessive time studying content that rarely appears in assessments while potentially overlooking frequently tested material. Traditional manual analyses of past exam papers and lecture notes to address these issues tend to be inconsistent, subjective, and labor-intensive, thus limiting their effectiveness.

In recent years, advancements in artificial intelligence, machine learning, and educational analytics have provided significant opportunities to automate and objectively evaluate educational content. Despite these technological advances, current learning management systems (LMS) utilized by universities typically lack robust analytical tools for systematic curriculum evaluation and improvement. Consequently, institutions face difficulty making informed, data-driven decisions about curricular changes.

Additionally, widely used revision materials and student-support resources often adopt a generic, one-size-fits-all approach, providing broad, syllabus-wide summaries rather than targeted, personalized study aids. This general approach can limit the effectiveness of revision and hinder the optimization of student academic performance by failing to reflect the actual trends observed in historical exam patterns.

To address these challenges, this research proposes an integrative, intelligent framework leveraging advanced Natural Language Processing (NLP) and educational analytics techniques. The goal is to systematically analyze historical exam questions, lecture content, and continuous assessments to classify, prioritize, and predict relevant topics, question difficulty, and curricular redundancies or gaps. Insights gained from these analyses will directly inform the development of personalized learning tools, including realistic model exam papers, concise short notes, adaptive essay feedback, tailored flashcards, voice notes, and individualized study schedules.

This integrative, multi-modal approach will employ transformative AI technologies such as transformer-based NLP models, semantic embedding methods, and automated content summarization techniques. Through this sophisticated blend of AI methodologies, the proposed research aims to deliver personalized educational support that accurately aligns with historical assessment trends and individual learner requirements. Ultimately, the project addresses the critical shortage of personalized, evidence-based, and data-driven academic support systems, enhancing learner engagement, curriculum effectiveness, and overall student achievement.

- Biggs, J., & Tang, C. (2011). *Teaching for Quality Learning at University* (4th ed.). McGraw-Hill Education (UK).
- Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81-112.
- Romero, C., López, M. I., & Luna, J. M. (2013). Predicting students' final performance from participation in on-line discussion forums. *Computers & Education*, 68, 458-472.
- Siemens, G., & Baker, R. S. (2012). Learning analytics and educational data mining: Towards communication and collaboration. *Proceedings of the 2nd International Conference on Learning Analytics and Knowledge (LAK'12)*, 252-254.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This research proposes the development of a comprehensive AI-powered educational support framework that enhances university students' self-learning, exam preparation, and assessment performance. The solution integrates Natural Language Processing (NLP), transformer-based models, rule-based logic, and personalized learning analytics to deliver targeted academic assistance.

1. Data Integration

- Collect historical exam papers, lecture materials, and continuous assessment (CA) questions.
- Use these academic data sources to understand past trends, topic coverage, and assessment patterns.

2. Intelligent Processing Pipeline

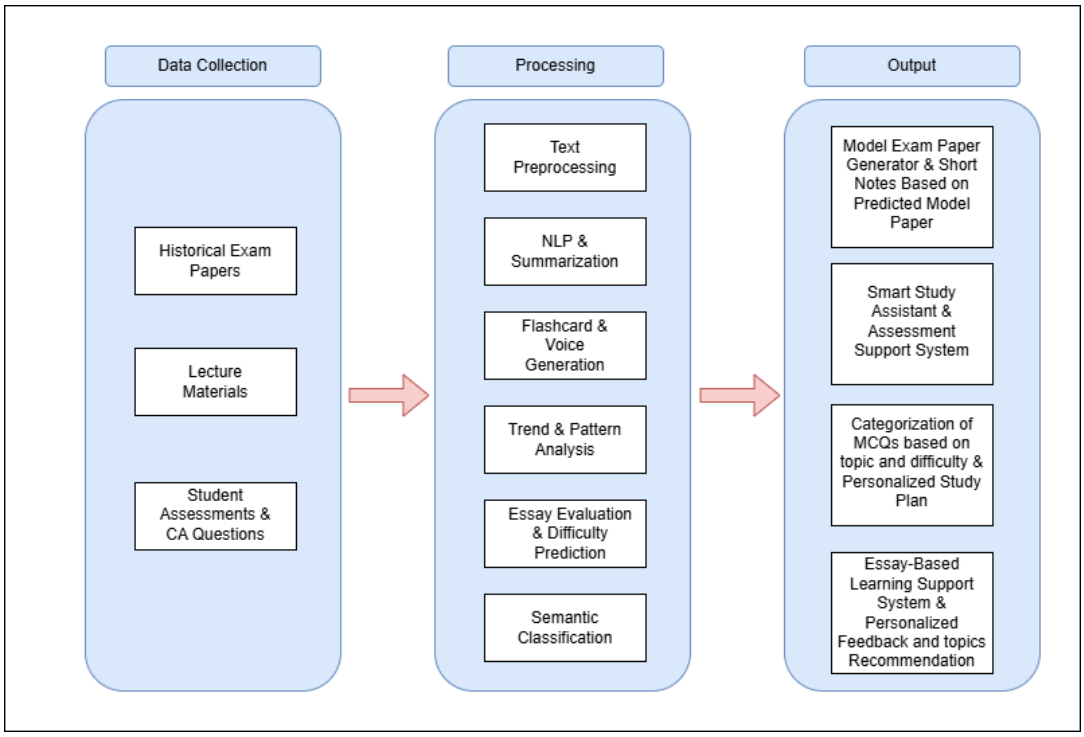
- Apply text preprocessing and transformer-based NLP models (e.g., T5, BART) to extract key information and generate structured summaries.
- Implement rule-based flashcard generation and Text-to-Speech (TTS) for multimodal content delivery.
- Perform trend and pattern analysis on historical exams to generate representative model papers.
- Use machine learning models to predict essay difficulty and provide semantic evaluation of student responses.
- Classify academic content into syllabus-aligned modules for study planning.

3. Personalized Learning Outputs

- Generate model exam papers and summary notes based on identified trends.
- Deliver an AI-powered smart study assistant that provides flashcards, voice notes, and step-by-step CA guidance.
- Categorize MCQs by topic and difficulty and generate personalized study plans based on performance and time constraints.
- Provide students with semantic-based essay evaluation and targeted feedback with suggested topics for revision.

4. Educational Impact

- Enhance student engagement, self-assessment, and revision effectiveness through adaptive, AI-driven content.
- Support educators by identifying difficult topics and common student weaknesses.
- Bridge the gap between lecture content, assessment standards, and individual learning needs through context-aware academic technologies.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Specialized Domain Expertise:

This project lies at the intersection of Artificial Intelligence, Data Science, and Educational Technology. It demands expertise in Natural Language Processing (NLP), machine learning, and educational analytics. Key areas include semantic analysis, curriculum alignment, and adaptive learning systems. A deep understanding of pedagogical frameworks such as Bloom's Taxonomy and cognitive-level tagging is essential for developing meaningful feedback and aligning AI-generated outputs with academic expectations and learner needs.

Knowledge:

The project requires proficiency in transformer-based NLP models like BERT, T5, and BART for text summarization, question classification, topic modeling, and difficulty prediction. It involves knowledge of both supervised and unsupervised machine learning techniques to support content classification and personalized learning recommendations. Practical skills in data preprocessing, content extraction, and semantic similarity analysis (e.g., using Sentence-BERT and cosine similarity) are vital. Additional competencies include rule-based flashcard generation, Text-to-Speech (TTS) conversion, and user interface development with tools such as Streamlit or React. Familiarity with educational data mining techniques supports trend analysis, topic prioritization, and study plan generation.

Data Requirements:

The research depends on diverse academic datasets, including 8–10 years of historical university exam papers, official lecture slides and notes, continuous assessment (CA) questions, and student-submitted essay responses. These datasets must undergo extraction, cleaning, normalization, tokenization, and semantic tagging to be processed effectively by NLP models. The quality, volume, and alignment of this data with the curriculum are critical for building, training, and evaluating the AI-driven academic support system to ensure relevance, accuracy, and educational impact.

8. Objectives and Novelty

Main Objective

The main objective of this research is to develop an intelligent, personalized, and data-driven academic support system that enhances student learning and exam preparation by analyzing historical exam papers, lecture content, and continuous assessments using advanced Natural Language Processing (NLP) techniques and educational analytics. This system collectively aims to classify and organize educational content, predict question difficulty, identify key topics, generate structured model papers and personalized short notes, provide adaptive essay feedback, and produce customized study plans. By integrating multiple AI-powered approaches such as semantic content mapping, transformer-based text summarization, difficulty indexing, adaptive feedback loops, and personalized recommendation systems, the project seeks to significantly improve students' efficiency, effectiveness, and engagement in their academic preparation, thereby bridging the gap between curriculum delivery, learner needs, and assessment expectations.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22891440	The primary aim of this component is to conduct an in-depth, data-driven analysis of	• **Collection of Academic Content:**- Gather and preprocess historical DMS exam papers (8–10 years).- Collect	The key innovation of this component is the automated and targeted creation of concise short notes, directly derived from lecture materials and precisely aligned with

	historical exam patterns within the DMS module, focusing explicitly on structural and topical trends. It seeks to produce a synthetic model exam paper that authentically reflects past exam formats, question types, and the most frequently tested topics. Further, it aims to create concise, highly relevant short notes extracted from lecture materials, carefully aligned to the topics in the generated model paper. This integrative process intends to enhance students' exam preparation	corresponding official lecture notes and slides. • **Exam Structure Analysis:**- Identify and document consistent structural patterns (sections, mark distributions, question formats). • **Topic Frequency Mining Using NLP:**- Extract high-frequency topics using NLP methods (e.g., TF-IDF, keyword extraction, semantic clustering).- Rank topics based on frequency to determine priority areas. • **Model Exam Paper Generation:**- Create a synthetic exam paper reflective of actual historical patterns, ensuring balanced topic distribution, realistic question styles, and appropriate mark allocation.3 • **Mapping Topics to Lecture Materials:**- Perform semantic and keyword-based matching to	a synthetically generated model exam paper. This approach significantly differs from traditional, broad-scope revision materials by offering narrowly tailored, high-impact revision resources exclusively focused on the most frequently tested topics identified through NLP-based exam analysis. By uniquely integrating detailed structural and topical analysis of past exams with semantic content mapping and intelligent summarization, this method demonstrates a novel application of data science techniques such as text mining, pattern recognition, semantic matching, and automated summarization. It highlights the practical value of data-driven approaches in educational contexts, providing students with personalized, efficient, and effective exam preparation support

	efficiency and effectiveness by providing tailored revision content derived from actual academic trends.	map model paper topics to relevant lecture content. • **Short Notes Generation:**- Generate concise and focused short notes per topic using extracted lecture material, ensuring clarity, depth, and direct relevance to potential exam questions. • **Output Validation and Review:**- Conduct thorough quality checks on the generated model paper and short notes.- Validate accuracy and practical utility through peer or mentor review.	
IT22171306	To develop an intelligent, AI-powered academic support system that enhances the self-learning process for university students by transforming lecture slides and continuous assessment (CA)	• **File Upload and Input Handling:** Develop a system that enables students to upload lecture slides or continuous assessment (CA) questions in various formats (PDF, DOCX, or plain text) and handles validation and secure storage for further processing. • **Text Extraction and Preprocessing:** Implement parsing logic using libraries like PyMuPDF and python-docx to extract raw text from uploaded	The novelty of this component lies in its integration of advanced artificial intelligence methodologies to develop a personalized academic support system designed specifically for university-level education. Unlike conventional e-learning tools, this system leverages transformer-based Natural Language Processing models to perform structured lecture summarization, employs rule-based logic for the automated generation of flashcards, and incorporates Text-

	questions into structured and personalized study materials. The system will utilize advanced Natural Language Processing (NLP) techniques and rule-based logic to automatically generate concise summaries, voice notes, and flashcards from lecture content. In addition, it will provide tailored step-by-step guidance for CA questions, particularly in the DMS domain, by using a fine-tuned Large Language Model (LLM) that is trained on historical university assignment data.	files, followed by text cleaning and formatting to ensure compatibility with downstream NLP tasks. • **Lecture Summarization using NLP:** Integrate transformer-based NLP models (e.g., T5 or BART) to generate structured and concise summaries from the extracted lecture content, with a focus on preserving slide-wise topic flow. • **Voice Note Generation:** Utilize a Text-to-Speech engine such as gTTS to convert each generated summary into an audio file, allowing students to review lecture content through voice playback. • **Flashcard Creation:** Design a rule-based flashcard generator that converts summarized content into question–answer pairs, structured to highlight key concepts and support active recall learning. • **CA Question Analysis:** Accept student-submitted CA or lab test questions specifically from the	to-Speech synthesis to produce audio representations of summarized content. This multi-modal approach enhances cognitive engagement by supporting both visual and auditory learning preferences. A distinctive feature of this system is its ability to analyze continuous assessment and laboratory test questions, particularly within the domain of Database Management Systems, using a fine-tuned Large Language Model that has been trained on institution-specific historical assessment data. As a result, the system can accurately interpret academic intent, deconstruct problem statements, and generate context-aware instructional guidance that includes recommended methodologies, tools, and reference resources. The holistic combination of content processing, intelligent assessment interpretation, and aligned academic recommendations within a unified platform represents a significant innovation in AI-assisted learning environments.

	This ensures that both content review and assessment preparation are closely aligned with actual university expectations and academic standards.	DMS domain and implement text processing to identify the core objectives and expected outcomes of each question. • **Step-by-Step Solution Guidance:** Fine-tune an existing language model on historical DMS CA questions to generate tailored step-by-step guidance, including method suggestions and relevant tools or approaches. • **Reference and Tutorial Link Generation:** Create logic to automatically recommend supporting material such as documentation links, academic references, and YouTube tutorial videos that align with the uploaded CA question topic. • **User Interface Development:** Build a responsive, user-friendly interface using a framework such as Streamlit or React to present summaries, flashcards, voice playback, and CA guidance in an organized and interactive format.	

IT22598196	This research focuses on Educational Technology, Artificial Intelligence (AI), and Natural Language Processing (NLP) to develop an intelligent essay-based learning support system. Educational Technology drives the creation of adaptive, personalized learning platforms. AI enables essay question difficulty prediction, response evaluation, and tailored feedback using machine learning. NLP facilitates semantic	Develop an intelligent essay-based learning support system with the following objectives: • **Analyze Materials**: Examine lecture materials and past exam data to predict essay question difficulty. • **Assign Difficulty Index**: Create a multi-level difficulty index for questions based on content traceability. • **Categorize Questions**: Organize past paper questions by difficulty level for student practice. • **Evaluate Responses**: Use NLP to assess student-submitted essay responses for quality, relevance, and completeness, applying Bloom's Taxonomy criteria. • **Provide Feedback**: Generate a satisfaction score and personalized feedback, including suggested topics for revision. • **Support Educators**: Offer insights into commonly	• **Unique Difficulty Prediction**: Integrates lecture materials and past exam data to generate a multi-level difficulty index based on content traceability, offering a novel approach to question complexity assessment. • **Advanced NLP Evaluation**: Employs NLP-driven semantic similarity and Bloom's Taxonomy criteria to evaluate essay responses, providing nuanced analysis of quality, relevance, and cognitive depth beyond traditional metrics. • **Adaptive Feedback Loop**: Delivers personalized feedback with targeted revision suggestions, fostering a dynamic, student-centered learning process. • **Dual Functionality**: Supports students with self-paced learning while providing

	analysis of lecture materials and student responses, supporting difficulty indexing and quality assessment. Related fields include Learning Analytics, identifying performance patterns; Adaptive Learning Systems, adjusting content to learner needs; Pedagogy, aligning with educational theories like Bloom's Taxonomy; and Cognitive Science, optimizing feedback for comprehension. This interdisciplinary approach enhances essay-based learning through	misunderstood topics to refine teaching strategies. • **Enhance Learning**: Foster self-paced learning and pedagogical improvement through an adaptive learning loop.	educators with data-driven insights into misunderstood topics, enabling pedagogical refinement. • **Subject-Agnostic Design**: Applicable across disciplines with essay-based assessments, enhancing scalability and versatility compared to domain-specific systems.

	predictive analytics and personalized pedagogical support.		

IT22113672	To semantically classify university exam questions and lecture materials into hierarchical syllabus modules using NLP techniques and to generate a time-aware, personalized study plan for students based on topic importance and individual preparation timelines.	• **Data Collection & Preprocessing:** Collect 10 years of past MCQ exam papers and corresponding lecture content for the selected subject module. The gathered textual data is then cleaned and normalized by removing irrelevant noise, tokenizing the text, and correcting any OCR-related errors to ensure consistency and readiness for further processing. • **Syllabus Module Classification:** Manually or semi-automatically label a portion of the questions and lecture content with their relevant syllabus module tags. Use Sentence-BERT to generate semantic embeddings that capture contextual meaning. A supervised classification model—such as fine-tuned BERT or a traditional classifier trained on these embeddings—is developed to classify the remaining data. The model is then validated and fine-tuned to	This component introduces a hybrid system that combines semantic classification using BERT with personalized time-aware study planning. Unlike traditional static study guides, the proposed method performs hierarchical syllabus mapping using contextual embeddings and incorporates student-specific constraints such as exam countdown and available study time. It leverages historical exam trends to prioritize topics and recommends targeted practice questions, offering a strategic, adaptive, and scalable exam preparation framework that bridges the gap between curriculum mining and personalized learning systems.

		achieve optimal classification accuracy. • **Topic Importance Analysis:** Using the classified questions, analyze the frequency and weight of each topic or module over the 10-year period. This statistical insight helps identify which topics appear more frequently, allowing the system to highlight high-priority areas based on historical exam trends and question density. • **Personalized Time-Aware Study Plan:** Collect input from individual students, such as their upcoming exam date and daily study hours. Based on topic importance and the number of days left for preparation, a scheduling algorithm allocates appropriate study time to each topic. The result is a personalized, topic-wise study schedule tailored to maximize preparation efficiency.	

		• **Practice Question Recommendation:** For each prioritized topic, retrieve semantically similar past exam questions using embedding-based similarity search or pre-indexed mappings. These targeted questions are recommended to the student to provide focused and relevant practice aligned with the personalized study plan. • **User Interface Development:** Develop a user-friendly and interactive interface that allows students to input exam details, view their personalized study plans, and access topic-wise practice questions. Enhance engagement through visual tools such as countdown timers, topic priority graphs, and study progress indicators, enabling students to track their preparation journey effectively.	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22891440	This component strongly aligns with the Data Science specialization by applying fundamental and advanced techniques in Natural Language Processing (NLP), text mining, pattern recognition, and automated content generation. It begins by systematically collecting and preprocessing historical DMS exam papers and lecture materials, employing structured parsing methods to extract valuable insights about exam formats, mark distributions, and question styles. Using NLP methodologies, such as keyword extraction (TF-IDF, RAKE) and semantic topic clustering (K-means, LDA), the component identifies the most frequently assessed topics within past exam papers. These analytical insights directly inform the generation of a realistic model exam paper, carefully constructed to mirror the historical patterns observed in the actual assessments. Subsequently, advanced semantic matching techniques (such as cosine similarity and Sentence-BERT embeddings) are utilized to map the topics featured in the model paper to corresponding sections within the lecture materials. Based on this precise semantic mapping, concise and topic-focused short notes are generated using extractive and abstractive summarization techniques. The novelty of this component primarily stems from this innovative automated generation of short notes that are strictly aligned with a synthetically generated model paper. Unlike conventional generic study resources, these short notes are narrowly tailored to students' revision needs, making the overall revision process efficient, personalized, and data driven. By leveraging diverse data science methodologies, this component effectively transforms unstructured educational content into actionable, student-centric learning support tools, demonstrating practical application and significant contribution to the field of educational analytics within the Data Science specialization.
IT22171306	This component is strongly aligned with the core competencies of the Artificial Intelligence and Data Science specialization, as it incorporates a wide range of AI-driven techniques, machine learning applications, and intelligent content processing strategies. The component uses Natural Language Processing (NLP) for lecture content summarization through transformer-based models such as T5 or BART, demonstrating an applied understanding of deep learning models for text generation and comprehension. The integration of Text-to-Speech (TTS) systems like gTTS further extends the use of AI by converting textual information into audio, supporting inclusive and multi-modal learning.

	From a data science perspective, the project includes the implementation of data extraction and preprocessing pipelines, leveraging tools such as PyMuPDF and python-docx, which reflects practical experience with unstructured data handling. The rule-based logic used for automatic flashcard generation represents symbolic AI, while the fine-tuning of a Large Language Model (LLM) using historical DMS continuous assessment data showcases supervised learning, model adaptation, and domain-specific natural language understanding. This enables the system to generate personalized step-by-step guidance for CA questions, demonstrating the ability to build intelligent, context-aware decision support systems. Moreover, the system's architecture is designed to provide recommendation capabilities through the automatic generation of academic references and tutorial video suggestions, highlighting the use of content-based filtering logic and semantic matching. The development of a user-friendly interface using frameworks such as Streamlit or React ensures that the component is accessible and practical for real-world educational use. Altogether, the component embodies key principles of the specialization by merging artificial intelligence, machine learning, educational data processing, and user-focused software development to create a comprehensive academic support system.
IT22598196	This component focuses on enhancing student learning through an intelligent essay-based learning support system that predicts essay question difficulty and provides personalized feedback. It begins by collecting lecture materials and past exam papers to form the foundation for analysis. These materials are analyzed to determine how directly answers to essay questions can be derived from course content, enabling the assignment of a multi-level difficulty index to each question. Next, Natural Language Processing (NLP) techniques, including semantic similarity analysis and topic modeling, are employed to categorize past paper questions by difficulty and content relevance. This analysis supports the creation of a practice environment where students can engage with questions tailored to their learning needs. Students submit essay responses, which are evaluated using NLP for semantic alignment with expected answers and Bloom’s Taxonomy criteria to assess cognitive depth, resulting in a satisfaction score and personalized feedback. The feedback provided includes specific suggestions for improvement, such as topics to revisit, and is designed to foster an adaptive learning loop that

	supports self-paced study. Additionally, the system aggregates student performance data to identify commonly misunderstood topics, offering educators actionable insights to refine teaching strategies. This component aligns with the Educational Technology and Artificial Intelligence domains by leveraging NLP, machine learning, and learning analytics. The novelty lies in the integration of difficulty prediction, semantic evaluation, and adaptive feedback within a single platform. By transforming lecture materials and exam data into a dynamic learning tool, this approach demonstrates how AI-driven methods can enhance personalized academic support and pedagogical improvement.
IT22113672	This component focuses on developing an intelligent system that leverages Natural Language Processing (NLP) and adaptive learning techniques to assist university students in exam preparation. It automatically classifies past MCQ exam questions and lecture materials into hierarchical syllabus modules using semantic embeddings generated by models like Sentence-BERT. The system begins by collecting and preprocessing 10 years of historical exam questions and lecture content, cleaning and normalizing the textual data for accurate analysis. A subset of this data is manually labeled with syllabus module tags to train a supervised classifier, such as a fine-tuned BERT model or other traditional classifiers, enabling the automatic categorization of new content according to syllabus structure. Following classification, the component performs an analysis of topic importance by measuring question frequency across syllabus modules over the years. This analysis helps identify high-priority topics based on historical exam trends. Student-specific inputs, including exam date and daily study time, are then incorporated to create a personalized, time-aware study schedule. A scheduling algorithm allocates study time proportionally to topic importance and the student's available preparation time, helping them prioritize effectively. To enhance practical learning, the component retrieves and recommends relevant past exam questions for each important topic using semantic similarity search or pre-indexed mappings.

	The system features a user-friendly interface that allows students to input exam details, view their customized study plan, and access recommended practice questions. Visual tools like countdown timers and topic priority charts improve engagement and self-monitoring. Overall, this component delivers a data-driven, personalized, and scalable solution that transforms traditional exam preparation into a strategic and efficient process.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr	Sanvitha	Kasthuriarachchi	
Co-Supervisor	Ms	Karthiga	Rajendran	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.